

Data test

Status & Methodology Report

What data exists, how it is tested, what has been found (32 hypotheses, 0 survived), and what is open next.

A deterministic, point-in-time research harness for Indian cash-equity strategies – rebuilt from scratch after its predecessor (market_gate) produced numbers that could not be trusted.

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Executive summary

Data test is a from-scratch rebuild of the market_gate research project, built to correct specific, measured failures in its predecessor: same-bar fills, no costs until the third session of work, a survivorship-biased universe, no benchmark comparison, and no running count of how many hypotheses had been tried. Five rules – deterministic, point-in-time, executable, benchmark-relative, and counted – are enforced in code, not by discipline alone.

As of today, **32 hypotheses have been tested with the same rigor**: 26 rejected outright, 6 accepted on their own training window but none survived independent (val) confirmation. **Zero for 32**. 13 of 32 beat their own placebo floor at all, and every one of those failed the very next check (a val re-test, or an honest-fills/cost re-simulation). The consistent pattern is not a coding bug – it is real, honestly measured effects that do not survive contact with real execution costs, real fills, and an honest placebo comparison, and specifically an entry-timing signature (buying right as a dislocation is detected, not before it).

The project is currently between phases: a large data-collection push (2026-08-23 to 2026-08-26) added roughly a dozen new category-specific sources – shareholding, insider trading, index reconstitution, ETF AUM, options chain, credit ratings, macro series, and, most recently, real per-stock mutual fund holdings from two AMCs (SBI and Axis) – and one new hypothesis (mf_accumulation) has been tested against the newest of these. This report inventories everything collected, explains the testing machinery precisely, summarizes what has been found, and lays out the open decision points for what comes next.

1. What this project is, and why the data was rebuilt

The predecessor project backtested 2004–2026 against 294 CSVs that are today's Nifty 500 members – a survivorship-biased universe. Measured on the one window where a comparison is possible (2019+), 22.5% of EQ symbols trading before 2020 are gone by 2026 (HDFC, MINDTREE, LTI, CADILAH, PVR, and 366 others); a survivor-only dataset shows 0% attrition there – the exact blind spot that matters. Auditing those CSVs also found two merged price series in 25 files, a trading calendar with 8,458 “days” over 26 years versus NSE's real ~6,650, rounding that destroyed six years of one stock's early returns, and literal Rs 0.00 closes.

Price history is therefore rebuilt from NSE's own daily cash bhavcopy archives – every traded symbol, no hindsight, back to 1995 – starting the usable window at 2004-01-01 on structural grounds (STT introduced Oct 2004; T+2 settlement replaced badla around 2001–02). The bhavcopy format also carries NSE's own corporate-action-adjusted *prev_close*, so a split or bonus is read from an exchange-published marker with an exact ratio, never guessed from a suspicious price jump.

The five rules, each enforced in code:

Deterministic. Same inputs, byte-identical outputs, provable via run manifests. An AST scan (*tests/test_determinism.py*) fails the build if any module reads the wall clock outside two allowlisted metadata timestamps.

Point-in-time. No datum reaches a decision before it existed. The universe is a recomputable rule (monthly rebalance, banded top-200-by-turnover), never today's index membership.

Executable. A signal from bar T's close fills at bar T+1's open, with real statutory costs and a participation cap. *config.execution.fill_at* raises on anything else.

Benchmark-relative. Every headline is excess return over NIFTY50, on capital, after costs – absolute return is never the reported number.

Counted. Every hypothesis tried is logged append-only to *hypothesis_log.csv*, win or lose, so the significance bar can rise honestly with the size of the search.

2. Data inventory – what exists today

Beyond the core price/F&O rebuild, this project has sourced 17 category-specific datasets itself, each read-only for the deterministic harness and each fetched via its own *scripts/fetch_*.py* – never a live call from inside a signal. Every source below was verified live before being trusted (real probes, not assumed availability), and every real limitation is stated plainly rather than smoothed over.

Category	Source	Coverage	Real caveat
Core	NSE cash bhavcopy (rebuilt)	2004–2026, every traded EQ symbol	Replaces predecessor's 294 survivorship-biased CSVs; 22.5% of pre-2020 symbols are gone by 2026
Core	F&O; bhavcopy (fno.db, read-only)	2008–2026	Pre-existing 48GB db; instrument-code taxonomy changed 2024-06, filter on <code>asset_class+contract_type</code>
Fundamentals	Quarterly financial results	~2007–2026, 706/926 symbols	Standalone only – no balance sheet in quarterly filings
Fundamentals	Shareholding pattern	~2021–2026, 597/926 symbols	Real NSE XBRL coverage floor, not a scrape limit
Flow	Insider trading (SEBI PIT)	2015–2026, 584/926 symbols, 251,933 disclosures	2015 = real regulatory floor
Flow	Index reconstitution calendar	2010–2026, 18,391 events, 303 indices	Pre-2010 NSE releases have no ticker column
Flow	ETF / Index-Fund AUM	2006–2026, 24,836 rows, 1,719 schemes	filing_date is an assumed 15-day lag
Flow	AMC portfolio disclosures (SBI + Axis)	SBI 2013–2026 (68 files); Axis 2021–2026 (3,672 files)	Raw workbooks; ICICI/HDFC scoped but need real browser automation
Flow	SBI + Axis per-stock equity holdings	Axis full range; SBI 2023–2026 only – 243,224 rows, 199 schemes	SBI's 2013-2016 files use a 6-col template not yet parsed
Derivatives	Per-stock options chain	2008–2026, 460 symbols, strike-level	Already in fno.db – raw only, IV/PCR/max-pain not built
Derivatives	Continuous front-month OI / futures price	OI from 2008; futures price from 2008	Two different stitching rules – OI can bridge a roll, price cannot
Credit/Rates	Credit rating actions	Apr 2023–2026, 1,080 actions, 6 agencies	Only 64.5% resolve to a real symbol
Credit/Rates	India G-Sec yields	2011–2026, 2 tenors (10Y, 3M)	Full curve needs an obfuscated RBI download, not built
Credit/Rates	Repo-rate proxy & bank credit	Repo 1968–2026; credit 1951–2025	Proxies, not the real RBI series – M3 dropped entirely, dead everywhere checked
Disclosure	Corporate announcements	2004–2026, 117,121 rows, 3,043 symbols	Best-behaved NSE feed found – both symbol and date genuinely filter
Disclosure	Earnings call transcripts	2010–2026, 19,045 full-text files, 1,436 symbols	~90% of volume is 2022+; 832MB text corpus
Macro	National macro series (CPI/II P/WPI/GDP/PLFS/forex)	Varies, 1990/1994–2026	MOSPI REST family; M3/repo/bank-credit not on this API
Macro	GST monthly collections	Jul 2017–Mar 2026, 105 months	Parsed from a GSTN retrospective PDF
Macro	Global cross-asset (VIX, crude, EM-FX, global equities)	Varies by series, through 2026-08	12 yfinance series; raw material, not wired into any signal

Investigated and explicitly not built

Item	Why
Bulk & block deal bulletins	Blocked – NSE's historical API returns a genuine server-side 503 on every parameter combination, confirmed reaching the real backend
Per-stock participant-wise OI	Closed – does not exist. NSE's own report is category × date only; no per-stock breakdown has ever been published
ICICI Prudential AMC portfolios	Deferred – confirmed to need real click-driven browser automation, each file URL only exists inside a React click handler
HDFC AMC portfolios	Deferred – data itself is trivial static links, but the site sits behind a real Akamai bot wall blocking all non-browser traffic
SBI legacy holdings template (2013–2016)	Scoped, unstarted – a genuinely different 6-column table, not yet parsed

A recurring, useful finding across this whole collection effort: the cost of getting a source rarely correlates with how important or large the underlying institution is. The largest mutual fund by AUM (SBI) and a much smaller one (Axis) both needed no browser automation at all; the 2nd- and 3rd-largest (ICICI, HDFC) were each genuinely expensive, for two completely different reasons (a JS-only click-driven UI, and a real bot wall). Each source was checked live rather than assumed from its category.

3. Analysis and backtesting mechanism

3.1 Architecture

config/config.toml every constant – a number in code is a bug
dtest/determinism.py seeds, content hashes, run manifests
dtest/config.py typed loader, validates on load (refuses same-bar fills)
dtest/universe.py point-in-time monthly rebalanced universe
dtest/data/ 17 source-specific fetch+parse modules (Section 2)
dtest/features/ point-in-time feature layer (technical, fundamentals, pairs, regime, MF holdings)
dtest/signals/ one file per hypothesis – 12 built
dtest/engine/ costs.py, futures_costs.py, simulate.py (long-only, T+1-open), pairs_simulate.py (two-leg), portfolio.py
dtest/evaluate/ metrics.py, placebo.py (30-seed), hypothesis_log.py (append-only)
scripts/ runnable entry points; every run writes a manifest
runs/ manifests + result CSVs, committed – the audit trail

3.2 The point-in-time universe

Rebalanced monthly (last trading day), 200 names by trailing 63-day turnover, with a 250-name band so an incumbent isn't dropped for a marginal rank slip (real turnover cost under this project's cost model). A name needs 252 days of history and to have traded within the last 5 sessions to be eligible at all. This is a recomputable rule applied identically to every historical month – never today's actual index membership projected backward.

3.3 Execution and costs

Every fill happens at the NEXT session's open after a signal closes – never same-bar. Delivery-equity round trip is **0.222% of position** on pure statutory rates (STT 0.1% each side, exchange transaction 0.00297% per side, stamp duty 0.015% buy-side, SEBI fee, 18% GST on brokerage+exchange+SEBI, zero brokerage – matches Zerodha delivery pricing), **0.322%** including the project's one named assumption, 5 bps/side slippage. Futures carry a separate, structurally different schedule (STT sell-side only, real non-zero brokerage). Rates are looked up from the real statutory schedule, not fitted.

Exit rule, shared across every signal so results stay comparable: a maximum 7-session hold, a 2.0×-ATR stop, and a 1:2.5 risk–reward target – whichever triggers first. Position sizing targets Rs 10,000 per trade at the trade level; the portfolio simulation separately runs a real Rs 50,000 account, 5 concurrent positions, equal-weight sizing, and a 40% max-sector-weight cap.

3.4 Statistical discipline

Two splits exist. **primary** (train 2004–2016, val 2017–2021, test 2022–2026-08) is the default, used by anything price-derived. **delivery** (train 2019-06-27–2023-06-30, val 2023-07–2025-03, test 2025-04–2026-08) is used by anything needing per-stock delivery, OI, participant-flow, or (as of this month) MF holdings data, which doesn't exist before 2019 – a real, stated higher evidence bar given roughly a third of primary's history. A 60-trading-day embargo separates every adjacent window.

Every real result is compared against **30 placebo seeds** – the identical signal count and timing, but firing on randomly chosen eligible names instead of the real selection rule – to establish a noise floor (6 seeds gave an empirical p of only ~0.14; 30 is the project's stated floor for a real claim). Significance is measured via a **non-overlapping (bucketed) t-statistic**, not a naive per-trade t-test, since trades from the same signal cluster in time and are not independent draws.

The rule that matters most for interpreting the results table below: **train decides, val confirms**. A candidate that looks good on train is never reported as a result until it has been re-tested, unmodified, on the held-out val window. Two constructions in this project's own history (same_sector_pairing, and the Oil & Gas-only cut of it) cleared t>2 with real statistical significance on train and were logged as “accepted” there – both then failed val outright. That is the discipline working as designed, not a failure worth second-guessing.

3.5 The hypothesis log

Every variant tried is appended to *runs/hypothesis_log.csv*, win or lose, and never edited or deleted – a later re-test of the same idea is a new row, not a rewrite. Each row requires a non-empty *story* field (the economic mechanism, stated before running anything) – the project's explicit defense against a parameter search masquerading as a hypothesis test. Rejections are logged with the same weight as acceptances; the file is committed to git like everything else, so “how many things have been tried” is answerable by reading a file, not by scrolling chat history.

4. Results so far

32 logged hypotheses, 26 rejected, 6 accepted-on-train-but-not-confirmed. Zero for 32. Thirteen distinct economic constructions have been tried, several across both splits and several parameter variants:

Construction	Idea	Log entries	Outcome
mean_reversion	50-day mean-reversion dip-buy	2	Only predecessor strategy that survived a walk-forward test – failed honest re-test on both splits
delivery_breakout	Price breakout confirmed by delivery %	1	Barely positive gross, below every placebo
oi_momentum	Price breakout confirmed by rising OI	2	Worse than every placebo on both splits
participant_tilt	Mean-reversion gated by FII flow regime	2	Directionally sensible, still below placebo
vol_squeeze_breakout	Breakout after a volatility contraction	3	Worst t-stat of any signal; an entry-delay variant improved but didn't save it
price_action (LONG)	Wide-range, volume-confirmed conviction bar	2	Confirms the entry-timing mechanism at its sharpest
pairs_reversion	Same-sector spread mean-reversion, market-neutral	3	Positive gross; collapsed once honestly costed
same_sector_pairing	Pure same-sector pairing, no correlation filter	6	Only construction to clear $t > 2$ on train – failed val both times
momentum (12-1 month)	Trailing 12-month momentum, skip-month	3	Accepted on delivery/train and val – never re-tested past val
earnings_surprise	SUE/PEAD quarterly earnings surprise	1	Positive gross, not significant
value	Trailing P/E vs own history	1	Positive t-stat, still below placebo
quality	TTM margin trend	1	No real signal either direction
mf_accumulation	Combined Axis+SBI MF holdings growth (2-AMC scope)	1	First non-price-derived mechanism tested – same entry-timing failure shape

The one consistent pattern across every rejection is not a coding bug – it is real, honestly-measured effects that do not survive contact with real execution costs, real fills, and an honest placebo comparison. A dedicated diagnostic (2026-08-15/17) confirmed the mechanism directly: every signal's real trades show a stop-hit rate meaningfully above their own placebo's, with target-hit rates flat or worse – each one buys right as a 1–2-day dislocation is detected, and the dislocation tends to continue for a session or two right into the fresh position before reverting. Delaying entry by 2 sessions measurably closed that gap on **vol_squeeze_breakout** (mean_net_pct –ve to marginally +ve) without producing a tradeable portfolio result – it removes the wrong reason to lose without yet producing a right reason to win.

The most recent entry, **mf_accumulation**, is the first test against a data source with no relationship to price, open interest, delivery, or participant flow – real per-stock mutual fund positioning (Axis + SBI's own combined holdings, 2 of ~50 AMCs by explicit scope decision). It failed the same way ($t = -2.94$, below its own placebo), which is itself informative: the entry-timing pattern generalizes beyond price-derived constructions, suggesting it is closer to a structural property of “buy right when a dislocation becomes visible” than an artifact of any one data transformation.

5. Further steps – open decision points

Nothing below has been decided yet; each is a real, live option rather than a queued task.

Extend the entry-timing-delay diagnostic to mf_accumulation

The one variant (delayed entry) that measurably helped a prior signal has not yet been tried on the newest one – a cheap, natural next check before concluding anything further about MF-holdings data.

Test the other two MF-holdings hypotheses scoped but not chosen

MF new-entrant (a scheme opening a brand-new position, a more discrete event than a gradual increase) and MF ownership breadth (a cross-sectional stability/quality read rather than a momentum-style trigger) were both scoped on 2026-08-26 and set aside in favor of mf_accumulation.

Build the SBI legacy-template holdings parser (2013–2016)

A real, scoped, unstarted job – a genuinely different 6-column table that would extend SBI's usable per-stock holdings history back from 2023 to 2013, subject to the already-known 2016–2023 filing gap.

Decide on HDFC / ICICI Prudential AMC data

Both confirmed to need real added investment (a bot-wall bypass, or real click-driven browser automation respectively) – worth it only if more AMC breadth is judged likely to change the mf_accumulation-style result materially.

Build a genuinely new construction on data already in hand

Not another parameter retune of a rejected construction. Real, unstarted candidates: a fundamentals trigger combined with the entry-timing delay already found real for technical signals; a cross-sectional fundamentals ranking rather than an absolute threshold; a signal built directly on shareholding/insider-trading/index-reconstitution data (promoter-pledge trend, insider net-buying streaks, pre-effective-date positioning ahead of a known index add) – none of these have been tried yet.

Retry the two blocked/deprioritized data items

Bulk & block deal bulletins (a real NSE-side 503, may have cleared); the 8 lower-priority macro series were already picked up 2026-08-26 and closed out.

This report is a snapshot, not a living document – regenerate it (*python scripts/build_project_status_report_pdf.py*) after any further data or hypothesis work to keep it current.